Software Requirements Specification (SRS)

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Contents

1 Introduction

The purpose of this application is to monitor and analyze urban development through AI-driven technologies, utilizing satellite imagery to detect unauthorized constructions and assess urban planning needs. The system aims to ensure compliance with city regulations, protect public resources, and support long-term sustainable urban development.

2 Problem Statement

In Pakistan, lack of city planning results in a haphazard urban expansion which contributes in creating future accommodation challenges and unequal access to essential services.

Unverified constructions further strain resources and disrupt sustainable development and hinder efficient city growth.

Our application makes use of AI-driven urban planning for the underdeveloped areas of the city to help build a more long term city plan. Moreover, it will make sure that the public resources are protected by keeping track of all the unverified and illegal constructions.

3 Scope

CitySentinel aims to provide a comprehensive solution for the monitoring of urban development as well as environmental conservation in Islamabad. It's core functionality is based on detection of illegal or unauthorized constructions and deforestation monitoring. This is done using satellite imagery and geographic data from Google Earth Engine and Google Maps API. Government authorities/officers will be able to input areas that are not permitted for construction and the system will detect any changes in these areas over time. In addition to this, green cover can be monitored, as can green area trends. This will help in assessing deforestation over time to help in sustainable environmental protection efforts.

The platform will include AI based urban planning, which will allow city planners to analyze and monitor land usage trends. This can in turn help them make informed decisions about future development projects. Users will have access to historical data about land usage, construction, and deforestation aiding them in identifying patterns and trends. The system will also provide notifications for illegal construction or environmental changes caused by deforestation, enhancing enforcement actions. This application will offer detailed data visualization tools and will display trends on the interactive map to simplify analysis.

The mobile app version will enable users to report illegal construction activity and receive alerts. Government officials will have access to a web based admin portal to manage areas of interest, analyze data, and then generate reports. There will also be integration of air quality and weather data to help assess the environmental impact the increasing urbanization has. The scope of **CitySentinel** focuses on sectors of Islamabad, helping authorities in better enforcing regulations while also helping preserve urban spaces.

4 Modules

Write down the modules of the proposed project. Each module should highlight features, using bulleted/numbered notation. When developing both a mobile app and a web app, group the modules according to the system types, such as, Client Web App, Client Mobile App, Admin Web App etc.

4.1 Module 1: Citizen Web App

The module lets users interact with the system using a web interface. It offers several features, including visualizing areas and reporting any illegal constructions.

1. User login and authentication.
2. Visualization of land usage and areas with greenery and deforestation.
3. Can alert upon detection of any of illegal constructions.

4.2 Module 2: Citizen Mobile App

A mobile based interface for the user. It has the same functionality as the citizen web app, but is optimized for mobile devices.

1. Login and authentication for users on a mobile device.
2. Push notifications for environmental alerts.
3. Visualization of land usage, deforestation and other key metrics.
4. Reporting illegal construction directly from the application.

4.3 Module 3: Professional User Web App

The module lets users interact with the system using a web interface. It offers several features, including visualizing areas, viewing changes to the environment, and monitoring construction activities. Users can also generate AI-driven city plans.

1. User login and authentication.
2. Interactive map interface used to take input for the area of interest.
3. City surveillance for the selected area.
4. Illegal construction monitoring.
5. Generate city plans using AI.
6. Urban development historical data.
7. Visualization of land use.
8. Visualization of areas with greenery and deforestation.
9. Alert upon detection of any of these illegal activities.

4.4 Module 4: Admin Web App

This module is specifically for government officials or any other authority. It will be used to configure monitoring areas, analyze urban data and then generate reports.

1. Admin login and verification/access control.
2. Input specific areas where construction is not permitted.
3. Monitor deforestation zones and green cover.
4. Analyze AI based urban planning suggestions.
5. Generate reports on illegal constructions and deforestation.

4.5 Module 5: AI-Driven Urban Planning

This backend module will be used for processing satellite images, detecting illegal construction and deforestation. It will also be used in analyzing trends.

1. Integrate Google Earth Engine for fetching and processing satellite images.
2. Detection of illegal constructions and deforestation according to the rules defined.
3. Storage and retrieval of satellite data for historical comparison.
4. Process data to monitor ongoing constructions.
5. AI based analysis for urban planning recommendations.

4.6 Module 6: Notifications & Reporting

The module is responsible for notifications and reports that will be generated for the users.

1. Generate alerts for illegal construction and deforestation.
2. Notifications for users (email, push notifications)
3. Generation and exporting of detailed reports of construction, environmental trends and system performance.
4. Scheduled reports for admins.

5 User Classes and Characteristics

User class	Description
Admin	Admin is the user with the most power over the system. This user can access all the functional features of the app such as AI-Driven urban planning, city surveillance, illegal construction detection, and check visualisations for land-use and plantation density. Apart from this, the admin will have access read and write to the database as well. Admin can create users, delete users and update user information. Moreover, the admin will receive notifications from the citizens as well. Moreover, admin can also generate reports that will include the latest information stored in the database regarding land usage and plantation density.
Professional Users	Professional users include the government officials and the city planners. These users will have access to the features: AI-Driven urban planning, city surveillance, illegal construction detection, and check visualisations for land-use and plantation density.
Citizens	Citizens will have access to the least features. They can simply just view land-use and plantation density visualisations. However, they can also report any illegal constructions spotted by them and send a notification to the admin.

6 Use Case Diagram

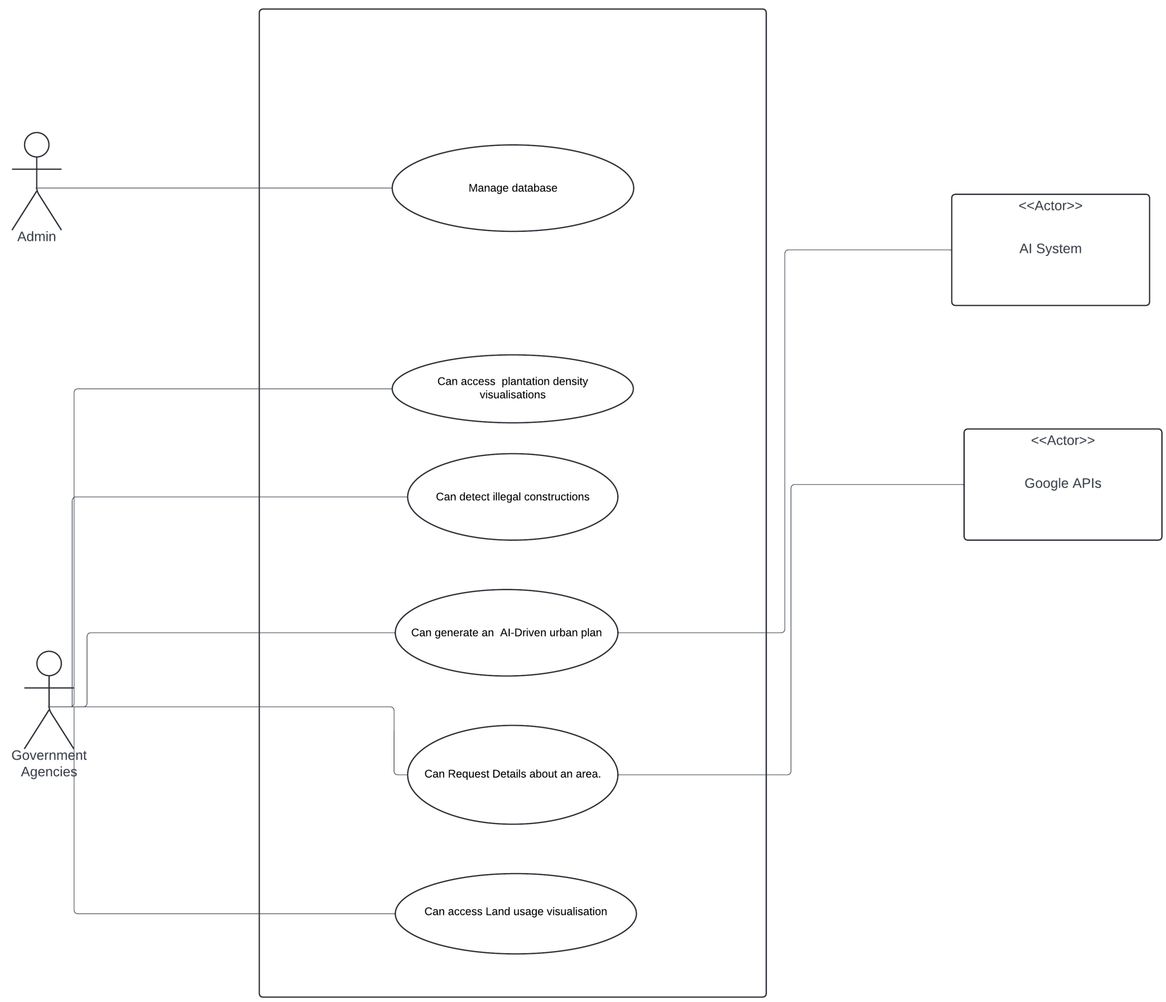

Figure 1: Use Case Diagram

7 High-Level Use Cases

Table 1: Manage Database

Use Case	Manage Database
Actors	Admin
Type	Primary
Description	The admin can perform CRUD operations on the database.

Table 2: Detect Illegal Constructions

Use Case	Detect Illegal Constructions
Actors	Government Authorities
Type	Primary
Description	They can select an area and receive information regarding the illegal constructions.

Table 3: Generate an AI-driven Urban Plan

Use Case	Generate an AI-driven Urban Plan
Actors	Government Authorities, AI System
Type	Primary, Secondary
Description	The user can select any underdeveloped area and get an AI-generated optimal city plan.

Table 4: Request City Details

Use Case	Request City details
Actors	Government Authorities
Type	Primary
Description	The user can get area details such as no. of houses, roads, schools etc, for the selected area.

Table 5: Access Plantation Density Visualisations

Use Case	Access Plantation Density Visualisations
Actors	Government Authorities
Type	Primary
Description	The user can get the plantation statistics of an area in forms of charts.

Table 6: Access Land-Use Visualisations

Use Case	Access Land-Use visualisations
Actors	Government Authorities
Type	Primary
Description	The user can get the details regarding land usage of an area in forms of charts.

8 Extended Use Cases

Table 7: Manage Database

Scope	The scope of database management operations.
Level	User Goal
Primary Actor	Admin
Stakeholder and Interests	Need to manage the database effectively.
Preconditions	Admin must be logged in.
Success Guarantee (Post Conditions)	Database operations are successfully performed.
Main Success Scenario	• Admin navigates to the database management page. • System displays database management options. • Admin selects a database operation. • System executes the operation and confirms success.
Extensions	System displays an error message for invalid input.
Special Requirements	Database operations must be secured.

Table 8: Detect Illegal Constructions

Scope	Detection of illegal constructions in an area.
Level	User Goal
Primary Actor	Government Authorities
Stakeholder and Interests	Need to identify illegal constructions.
Preconditions	Authorities must be logged in. Must give area input.
Success Guarantee (Post Conditions)	Illegal constructions are identified.
Main Success Scenario	• Authorities enter an area name. • System analyzes the area for illegal constructions. • Authorities review the analysis results. • System displays detected illegal constructions.
Extensions	System alerts if data for the area is missing.
Special Requirements	Analysis must be accurate to detect illegal constructions.

Table 9: Generate an AI-driven Urban Plan

Scope	Generate an optimal urban plan using AI.
Level	User Goal
Primary Actor	Government Authorities, AI System
Stakeholder and Interests	Need an optimal plan for urban planning.
Preconditions	Authorities must provide input data for the system to understand constraints.
Success Guarantee (Post Conditions)	An optimal urban plan is generated.
Main Success Scenario	• Authorities provide input data for planning. • AI system processes the data and generates a plan. • System displays the optimal urban plan.
Extensions	System alerts if input data quality is insufficient.
Special Requirements	The AI system must create an optimal plan.

Table 10: Request City Details

Scope	Request details about a city or area.
Level	User Goal
Primary Actor	Government Authorities
Stakeholder and Interests	Need detailed city information in a few clicks.
Preconditions	Authorities must be logged into the system. Input must be in the correct format.
Success Guarantee (Post Conditions)	Requested city details are provided in a reasonable time.
Main Success Scenario	• Authorities request details for a specific city. • System fetches the information using google APIs. • System will process the images. • System will retrieve information from the processed images. • System will display the data to the authorities.
Extensions	System notifies if the requested data is not available.
Special Requirements	Information provided must be accurate and up-to-date.

Table 11: Access Plantation Density Visualisations

Scope	Access visualizations for plantation density.
Level	User Goal
Primary Actor	Government Authorities
Stakeholder and Interests	Need to review plantation density data for environmental planning.
Preconditions	Authorities must be logged in.
Success Guarantee (Post Conditions)	Plantation density visualizations are accessible and displayed in forms of charts.
Main Success Scenario	• Authorities select the option ”plantation density visualization”. • System generates and displays the information in a form of visualizations.
Extensions	System notifies if there is an error generating the visualizations.
Special Requirements	Visualizations must be clear and informative.

Table 12: Access Land-Use Visualisations

Scope	Access visualizations for land use data.
Level	User Goal
Primary Actor	Government Authorities
Stakeholder and Interests	Need to review land use data for planning.
Preconditions	Authorities must be logged in.
Success Guarantee (Post Conditions)	Land use visualizations are provided.
Main Success Scenario	• Authorities select the option ”land use visualization”. • System generates and displays the data in forms of charts and graphs.
Extensions	System alerts if no data is available for the selected options.
Special Requirements	Visualizations must be accurate and up-to-date.

9 Functional Requirements

- **User Registration and Login:**
  - The system must allow users to register with their unique credentials.
  - Users must be able to log in using their credentials after registering successfully and then should be redirected to their role-specific dashboard.
  - Passwords must be encrypted for data security.
- **Role Based Access:**
  - The system must assign role-based permission and access for different user types.
  - Admin users must have full access to manage areas, view analytics, and generate reports.
- **Satellite Data Processing:**
  - The system must integrate Google Earth Engine and Google Maps API for retrieval and processing of satellite images.
  - The system must monitor designated sectors for changes and log these events.
- **Illegal Construction Detection:**
  - The system should allow users to define restricted construction zones via map inputs.
  - The system must analyze satellite imagery to detect unauthorized constructions and notify when any violation is detected.
  - The system must generate a report for violations, including information about the site.
- **Deforestation Monitoring:**
  - The system must track changes in green cover and identify areas with significant deforestation.
  - A visual representation of deforestation trends must be provided on the map, and reports must be generated for official use.
- **Push Notifications:**
  - The system must send push notifications to mobile and web users regarding illegal constructions or environmental violations.
- **Interactive Map Input:**
  - Govt official users must be able to specify regions on a map where construction or deforestation is prohibited.
  - The system must update these areas for future monitoring.
- **AI Driven Urban Planning:**
  - The system must provide AI-based analysis for optimal urban planning, based on current land usage and predicted growth trends.
  - AI recommendations must be updated as new data is processed.
- **Report Generation:**
  - The system must generate reports on illegal constructions and environmental changes.
  - Reports must be downloadable in PDF and CSV formats and accessible on web and mobile interfaces.
- **Mobile Application Access:**
  - The system must provide a mobile app for users to access key functionalities such as reporting, receiving notifications, and map viewing.

10 Non-Functional Requirements

Non-functional requirements include system performance metrics, security standards, and compliance with regulatory requirements.

- **Performance:**
  - For majority operations, the system must respond to user actions in less than two seconds.
  - The system should also be able to process large datasets of satellite imagery within a reasonable amount of time, e.g., 5 minutes.
- **Scalability:**
  - The system must be scalable in order to accommodate more users and wider monitoring areas.
  - It should be able to integrate of more datasets or APIs if needed.
- **Security:**
  - Transmission of data must be secure, making use of encryption protocols.
  - In accordance with data protection laws, all user information and data must be handled securely.
- **Availability:**
  - To ensure high availability, system up-time must be at least 99
- **Compliance:**
  - The system must be in compliance with Pakistan's construction and environmental regulations.
  - It must follow industry guidelines and standards for reporting and processing geographical data.
- **Maintainability:**
  - In order to facilitate future updates and maintenance, the system's code-base must be modular and well documented.
  - To maintain the system, and keep it current and secure, patches and updates must be installed on a regular basis.
- **Usability:**
  - The system must be user-friendly, requiring little training for both citizens and government officer users.
  - The usability and responsiveness of a design must be ensures on various platforms, including desktop and mobile ones.

11 Domain Model

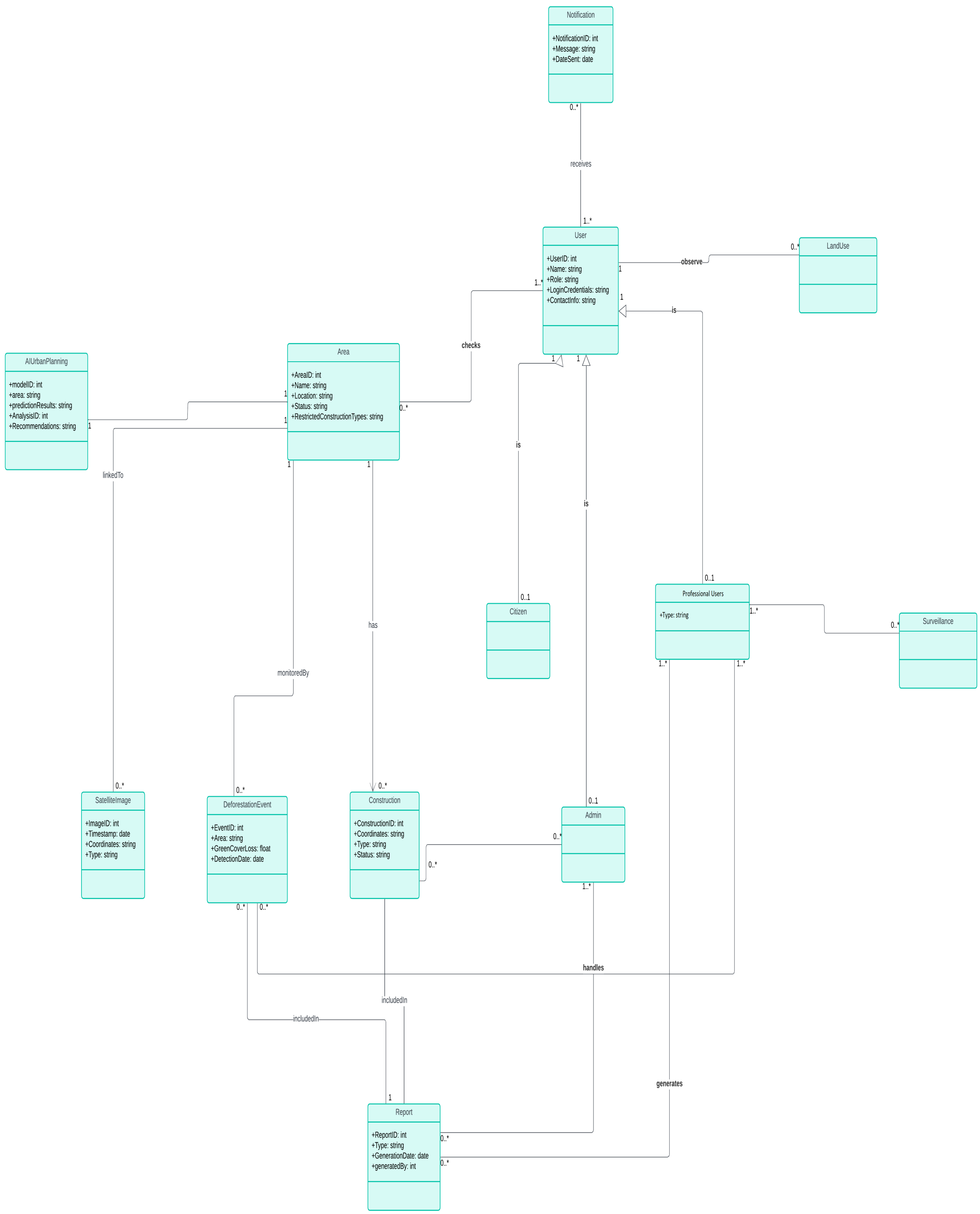

Figure 2: DomainModel

12 Design Models (Procedural Methodology)

12.1 Architecture Diagram

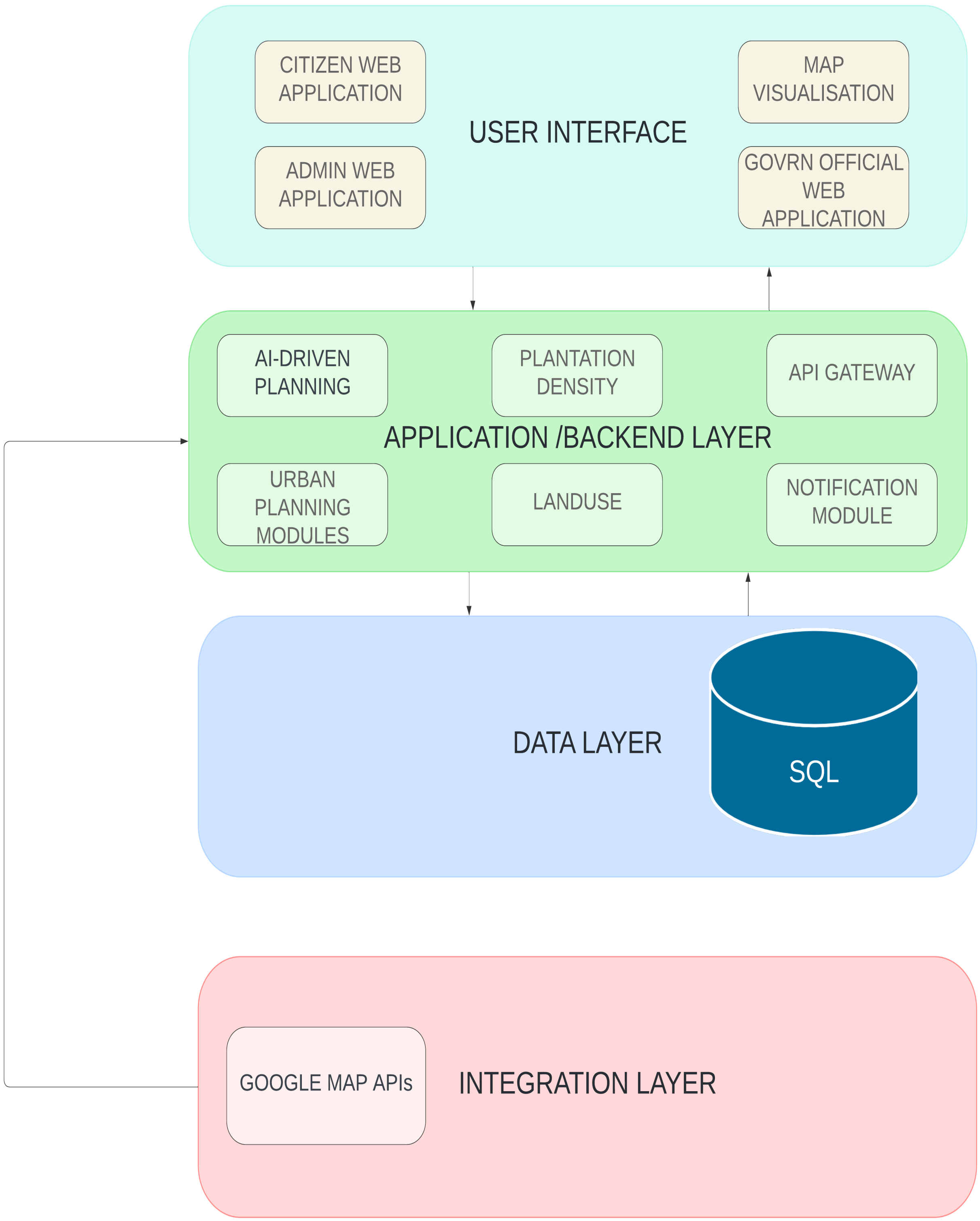

Figure 3: Architecture Diagram

12.2 Activity Diagram

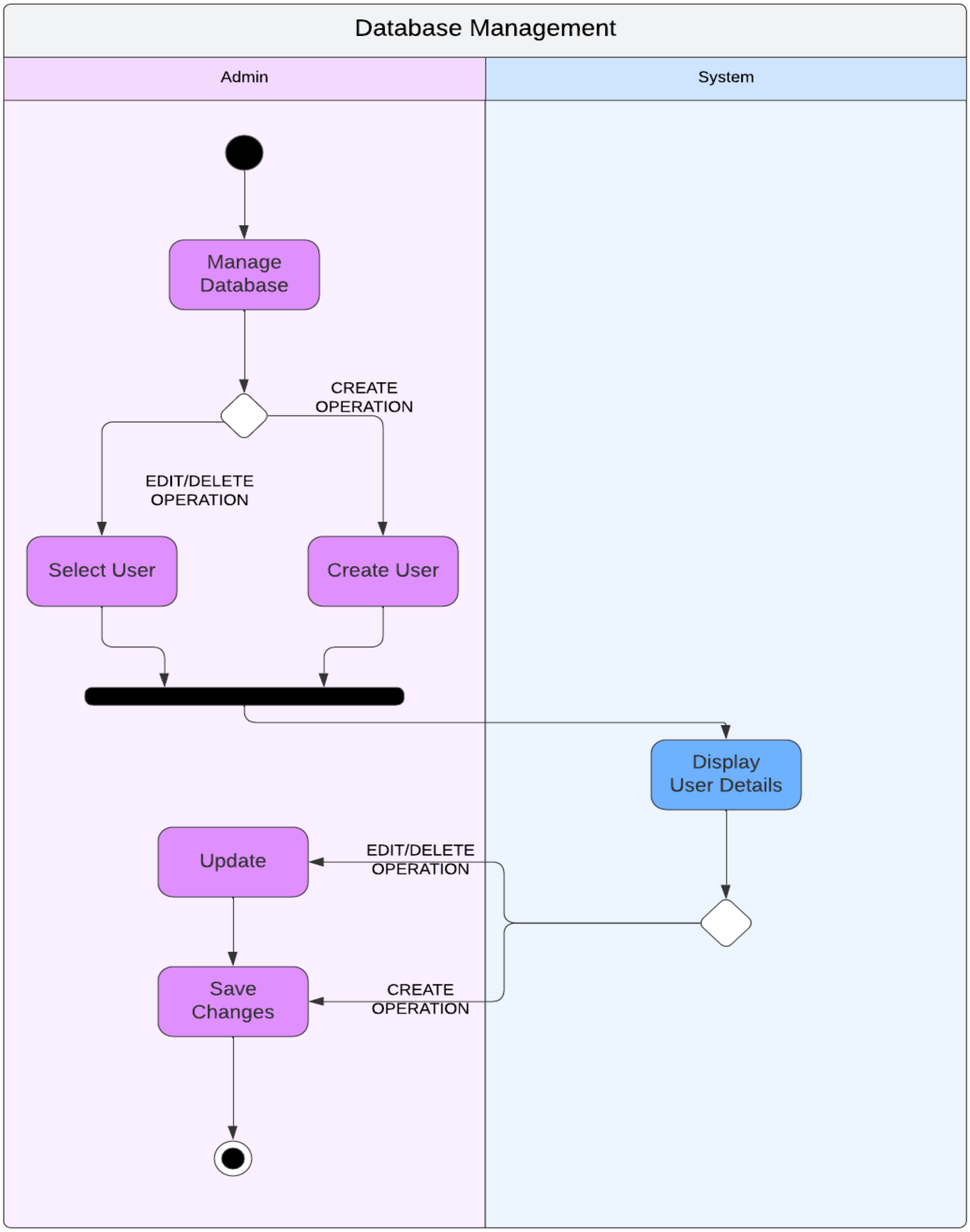

Figure 4: Use Case 1: Manage Database

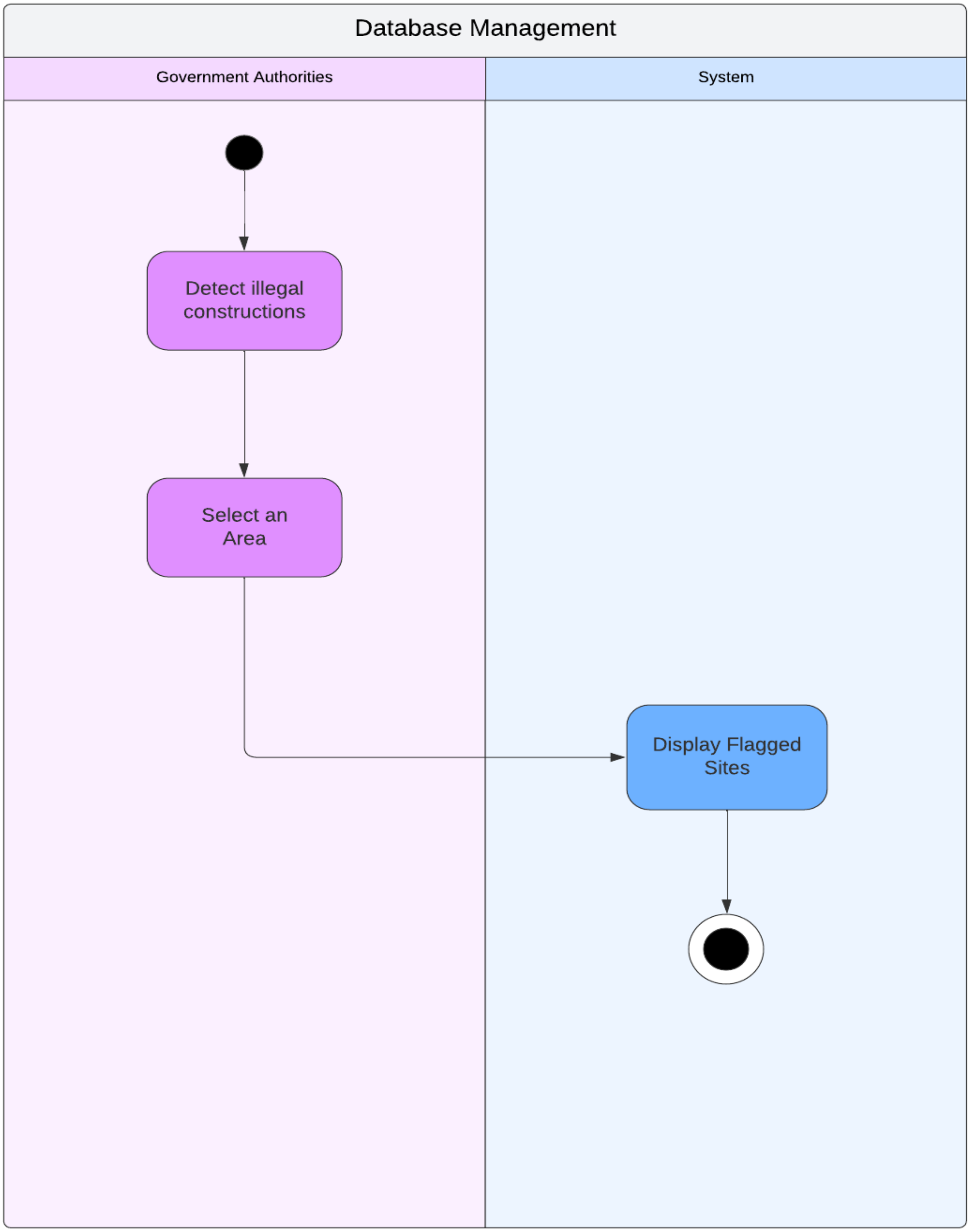

Figure 5: Use Case 2: Detect Illegal Constructions

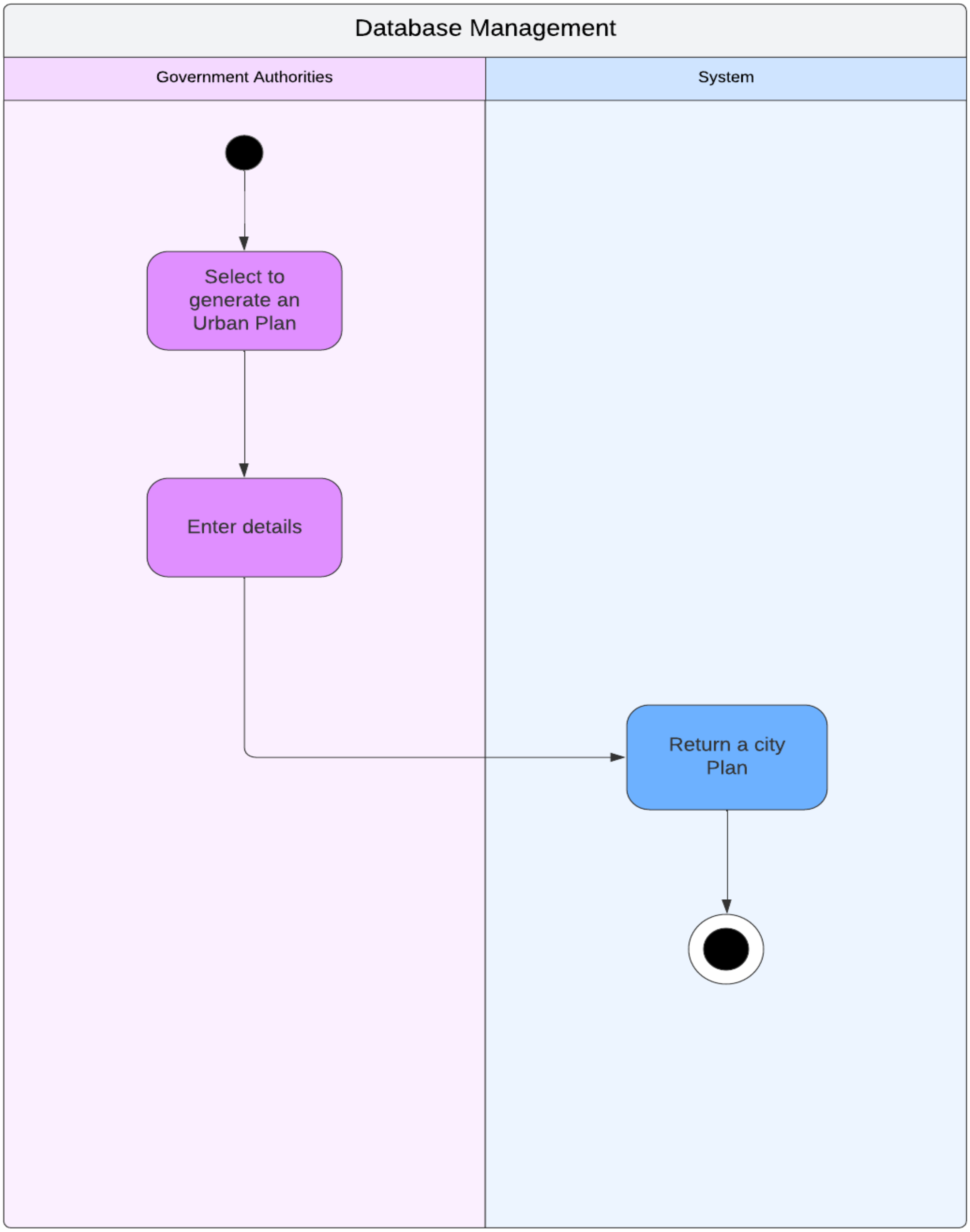

Figure 6: Use Case 3: Generate an AI-driven Urban Plan

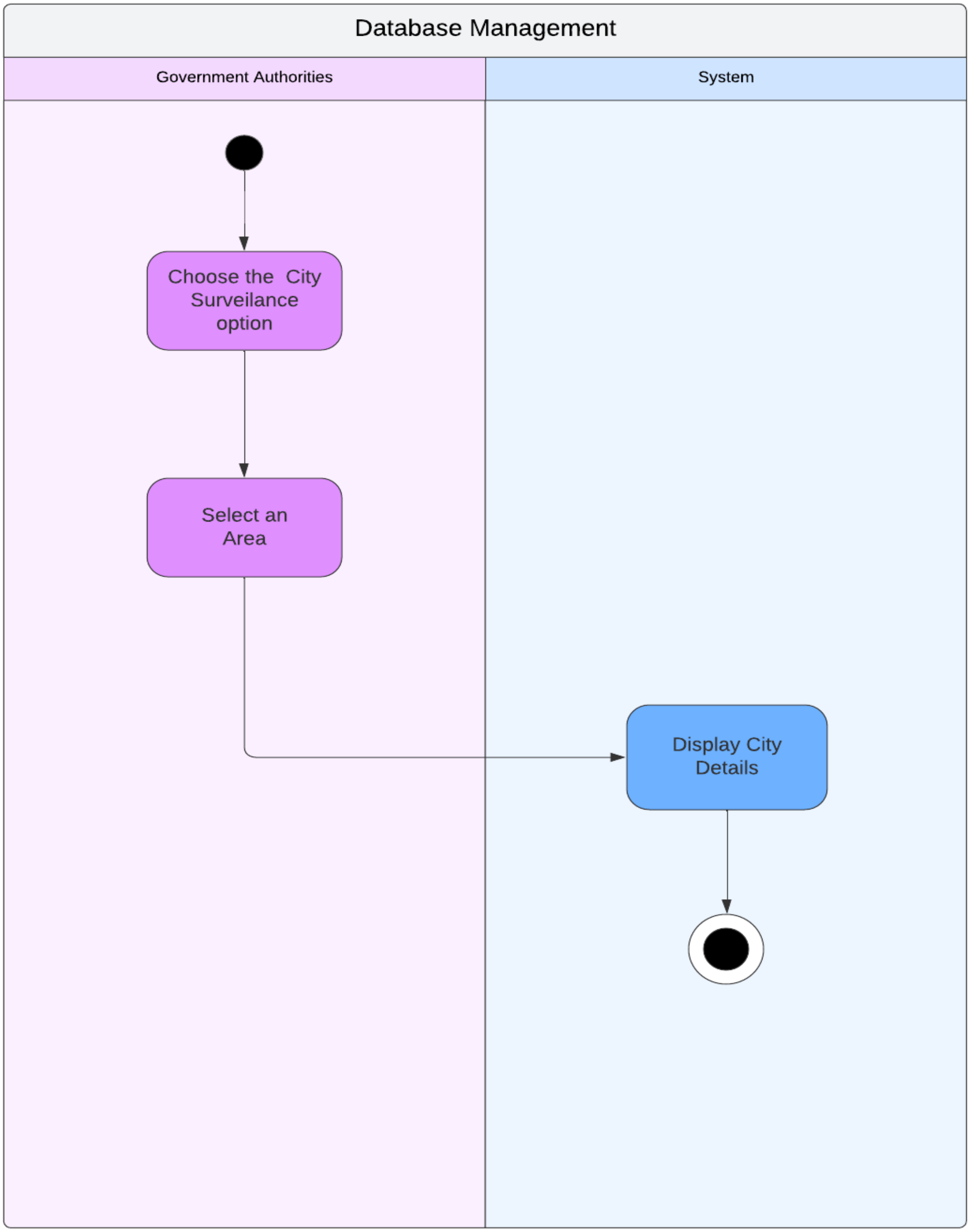

Figure 7: Use Case 4: Request City Details

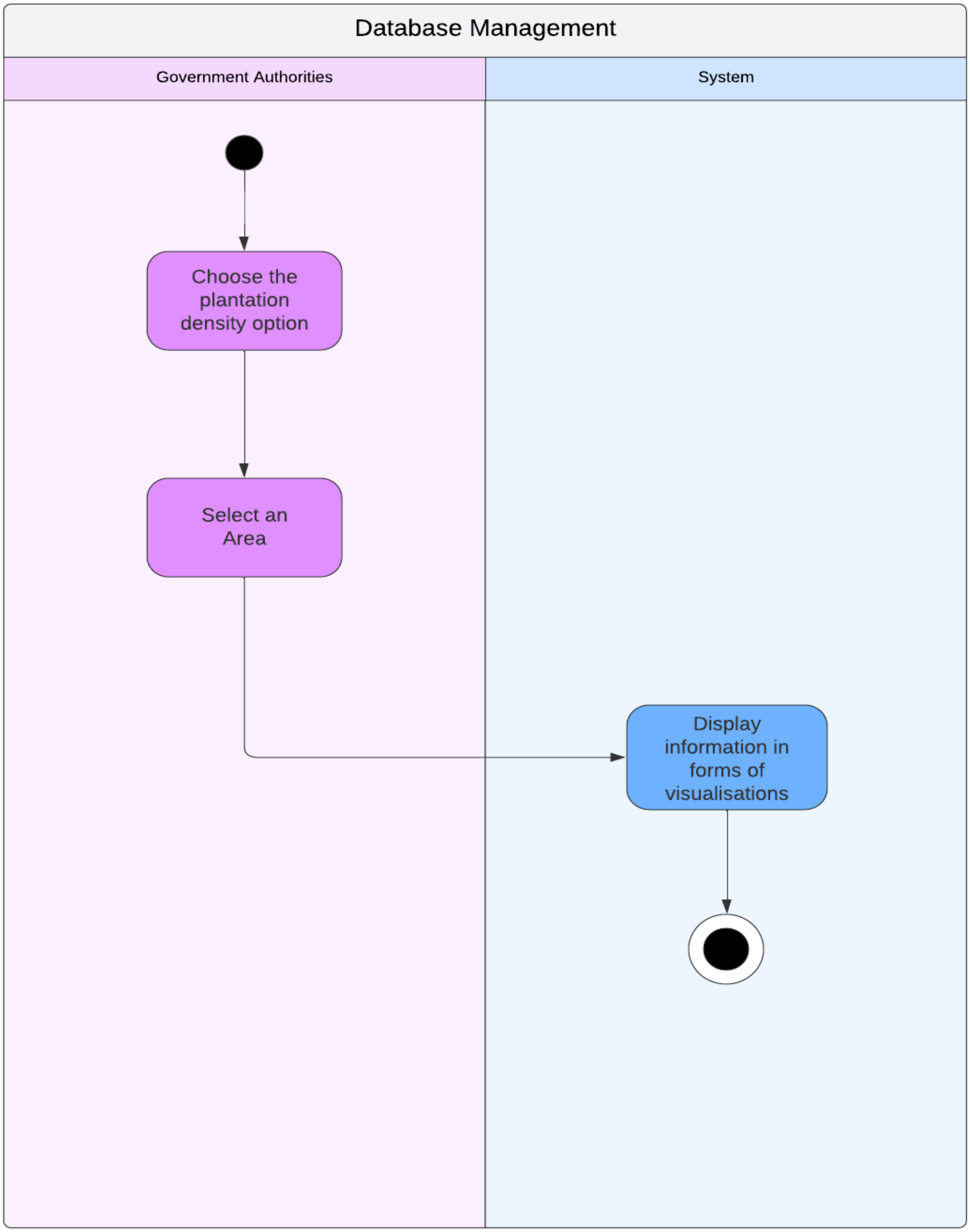

Figure 8: Use Case 5: Access Plantation Density Visualisations

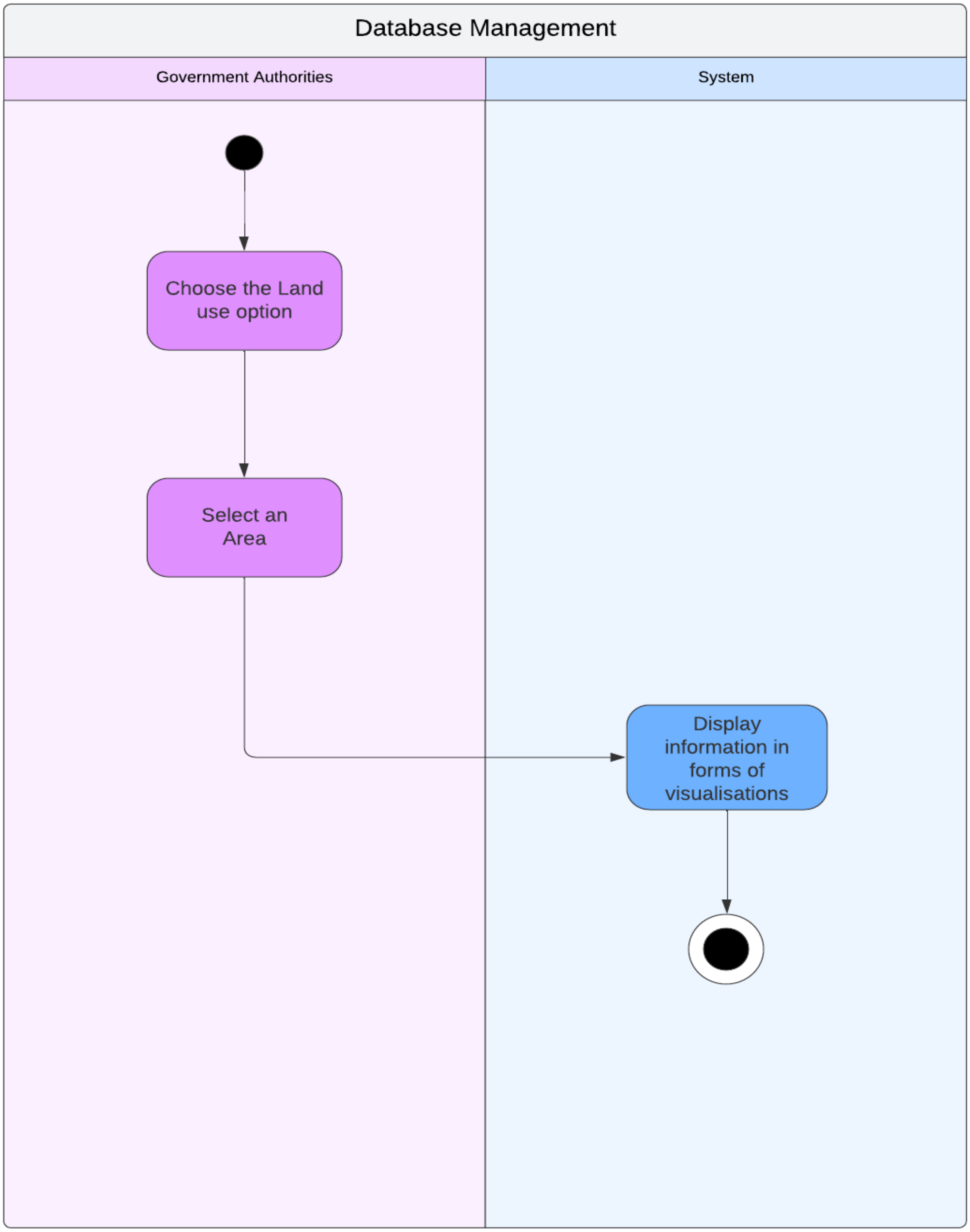

Figure 9: Use Case 6: Access Land-Use Visualisations

12.3 Class Diagram

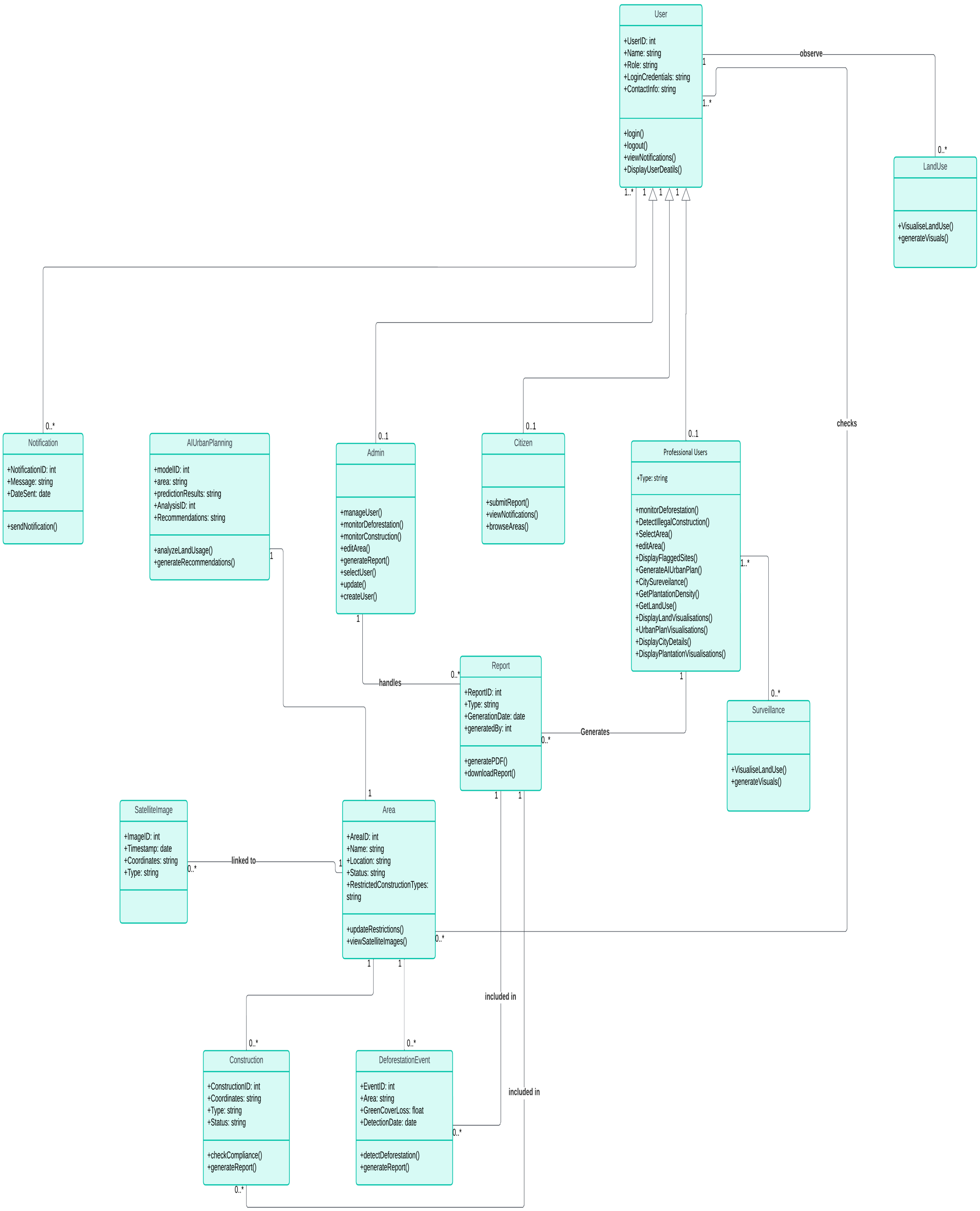

Figure 10: ClassDiagram

12.4 System Sequence Diagram – SSD

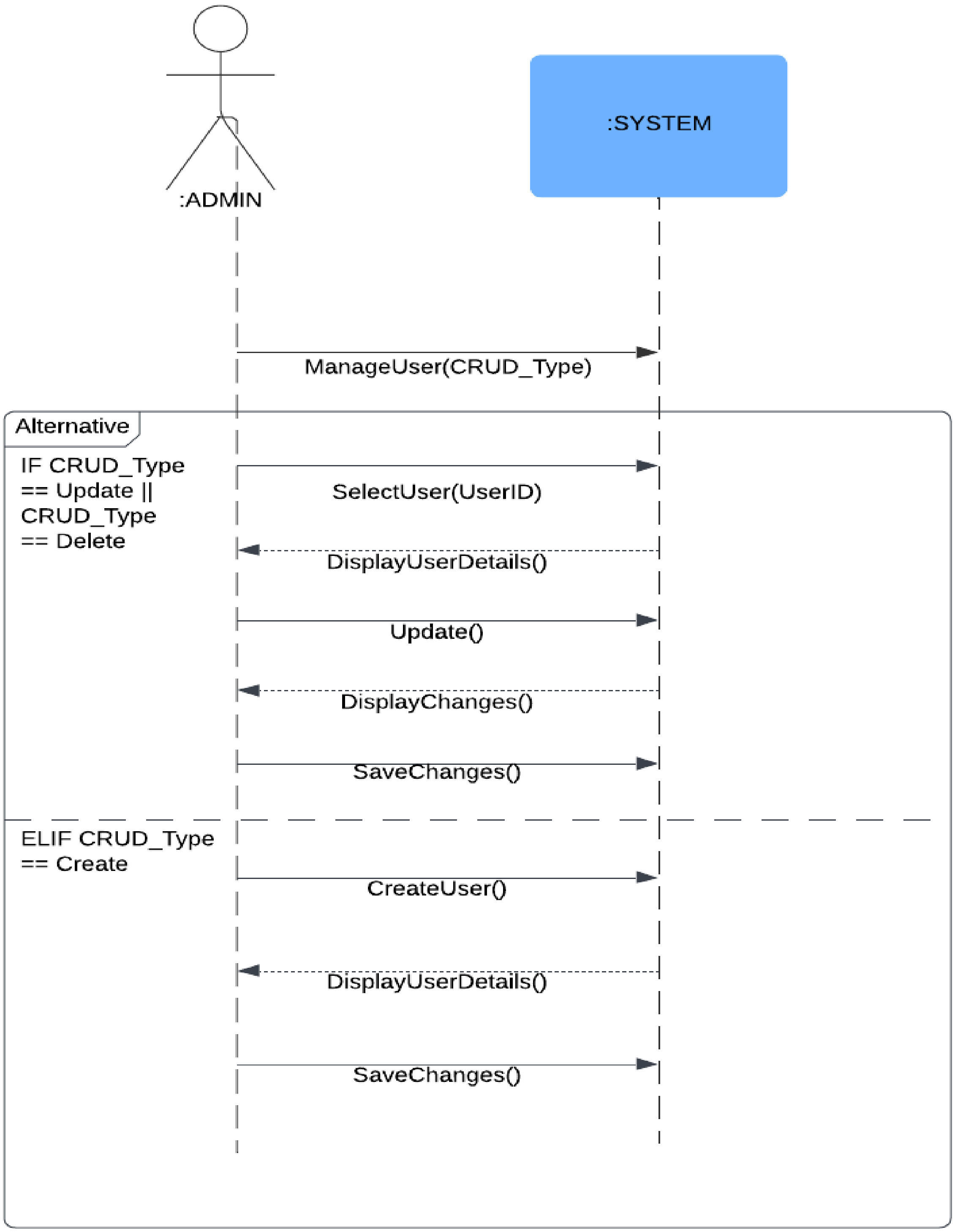

Figure 11: SSD for Use Case 1: Manage Database

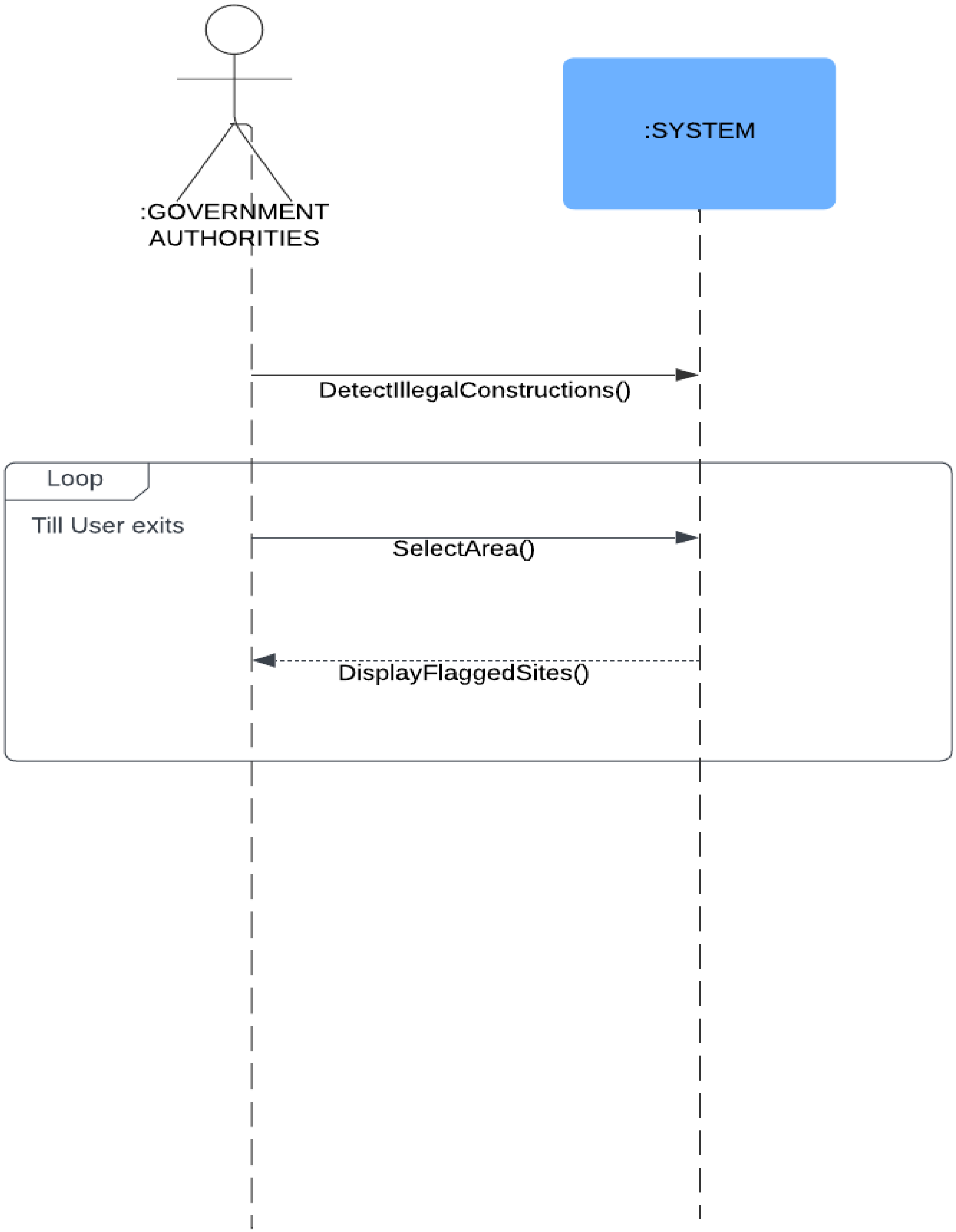

Figure 12: SSD for Use Case 2: Detect Illegal Constructions

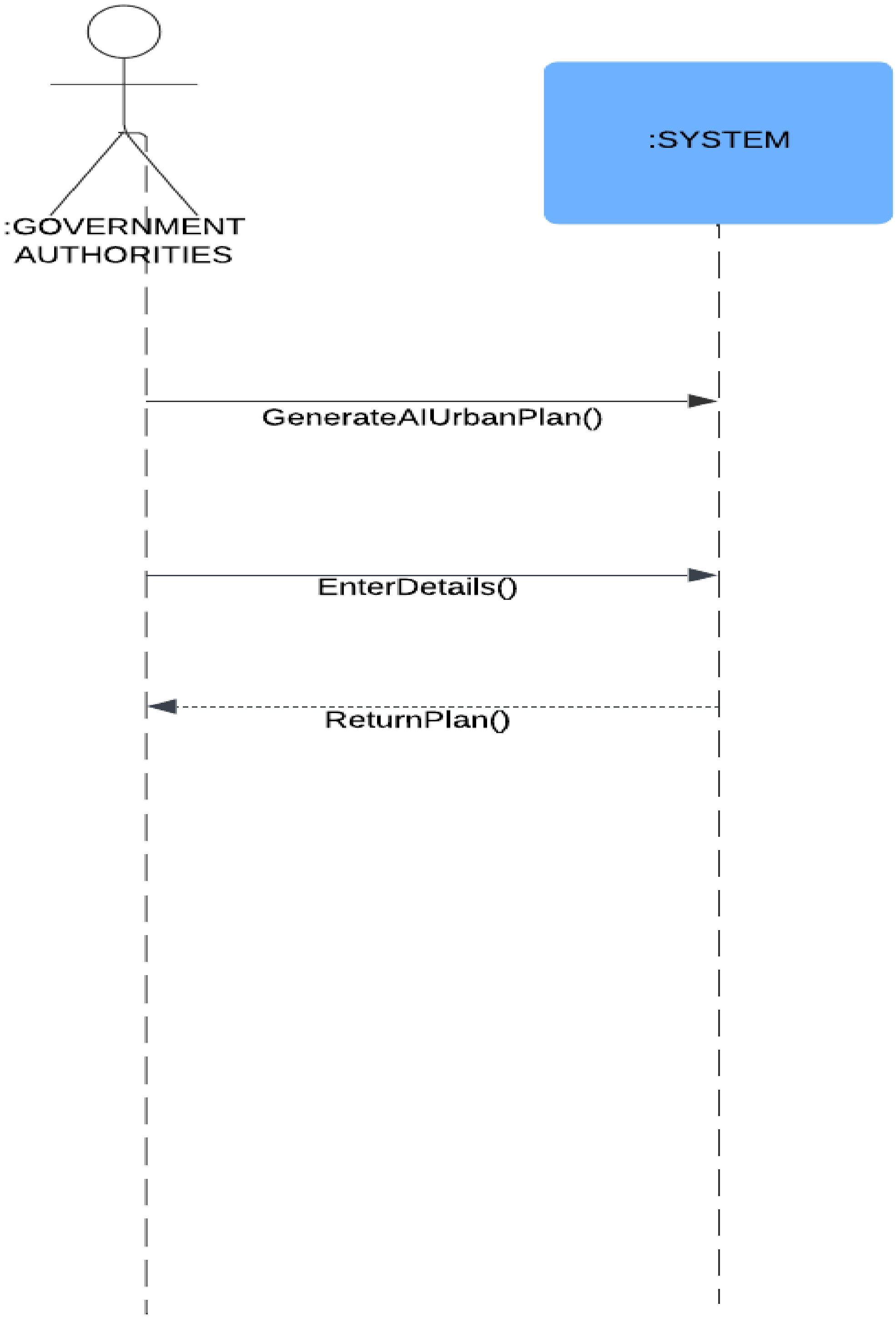

Figure 13: SSD for Use Case 3: Generate an AI-driven Urban Plan

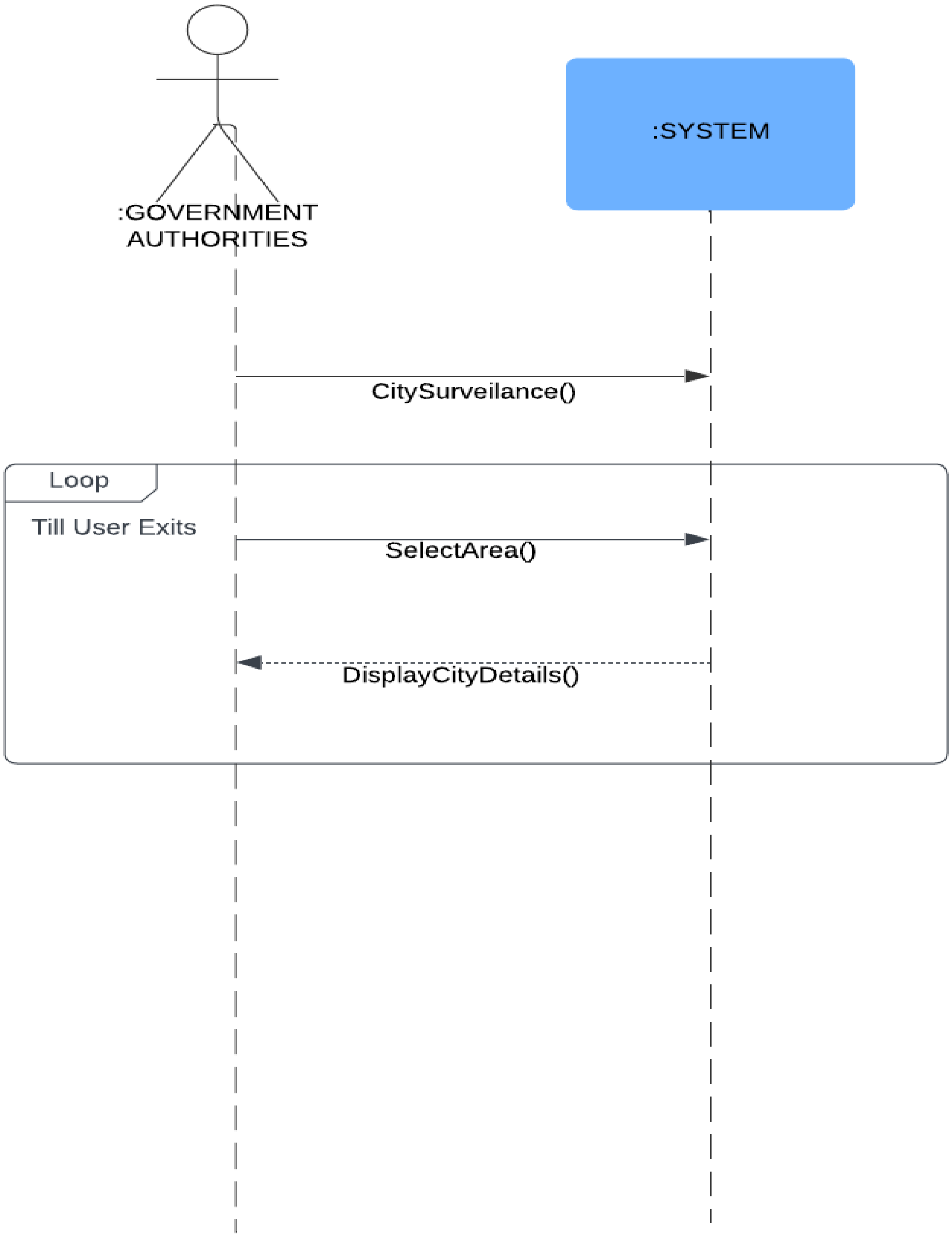

Figure 14: SSD for Use Case 4: Request City Details

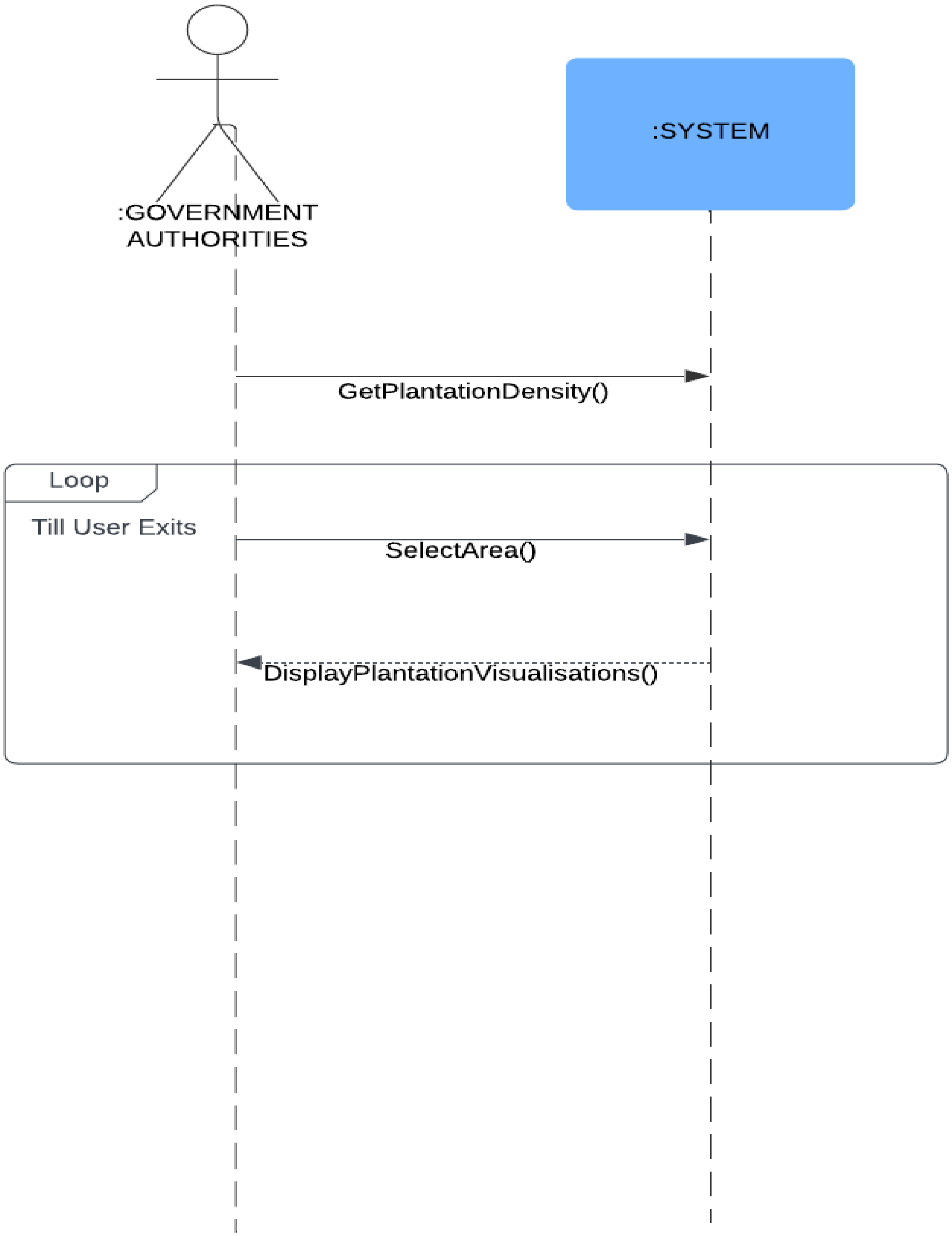

Figure 15: SSD for Use Case 5: Access Plantation Density Visualisations

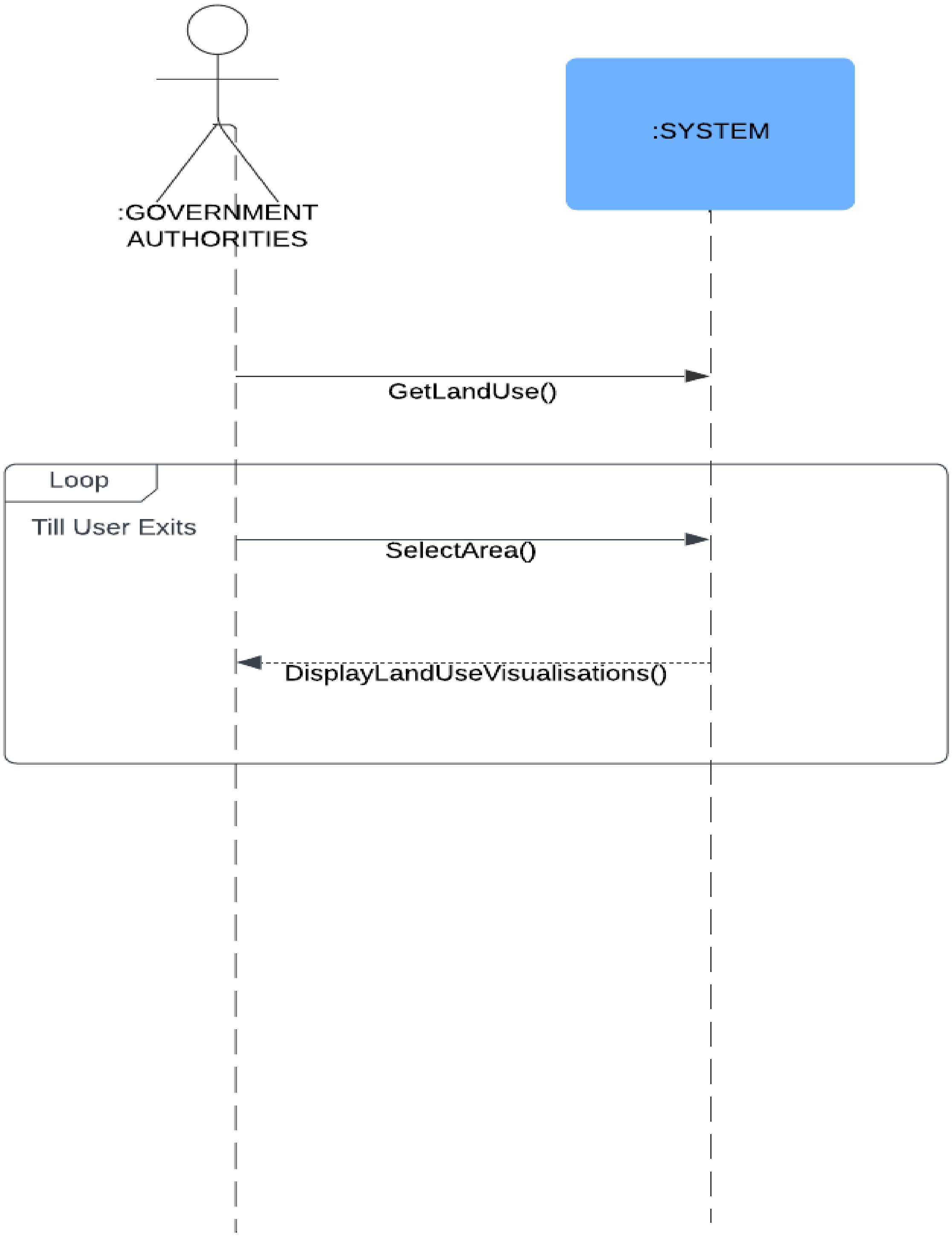

Figure 16: SSD for Use Case 6: Access Land-Use Visualisations

12.5 Sequence Diagram

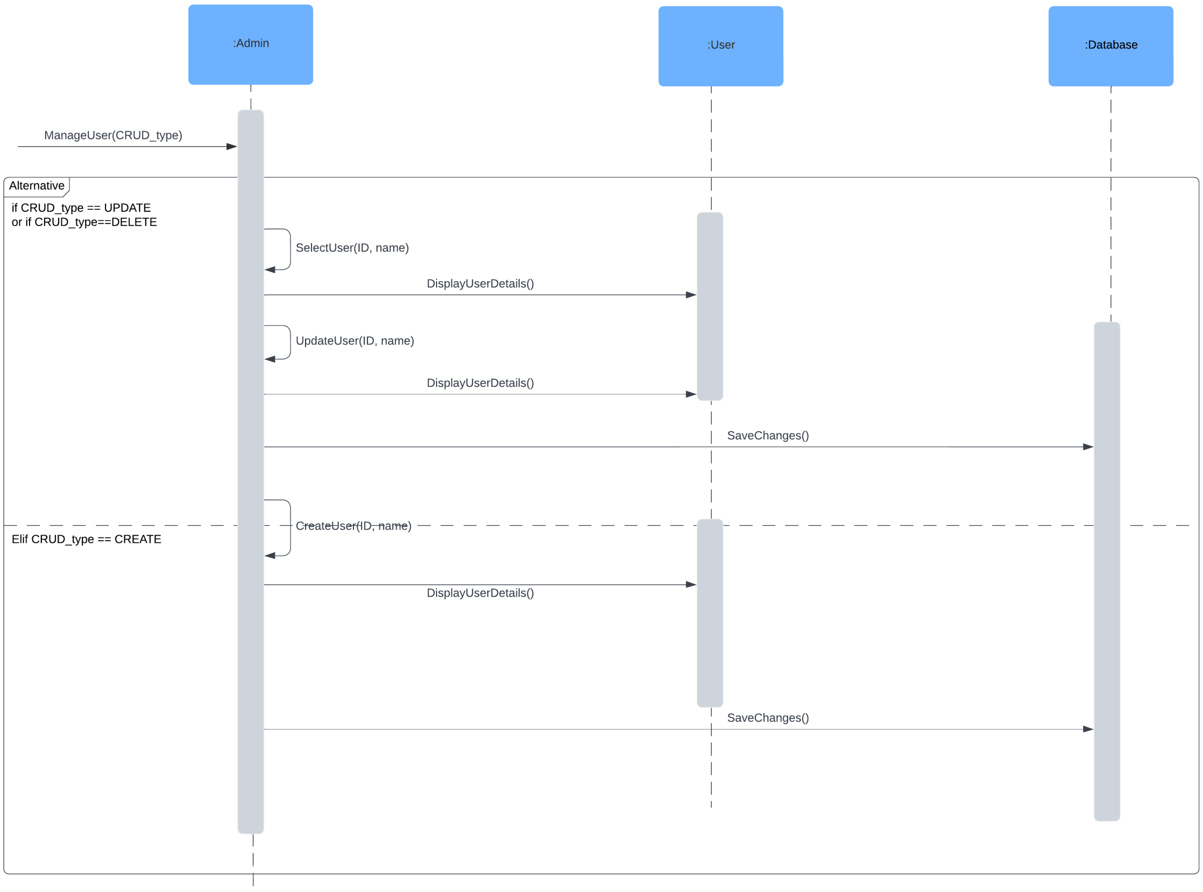

Figure 17: SD for UseCase 1: Mange Database

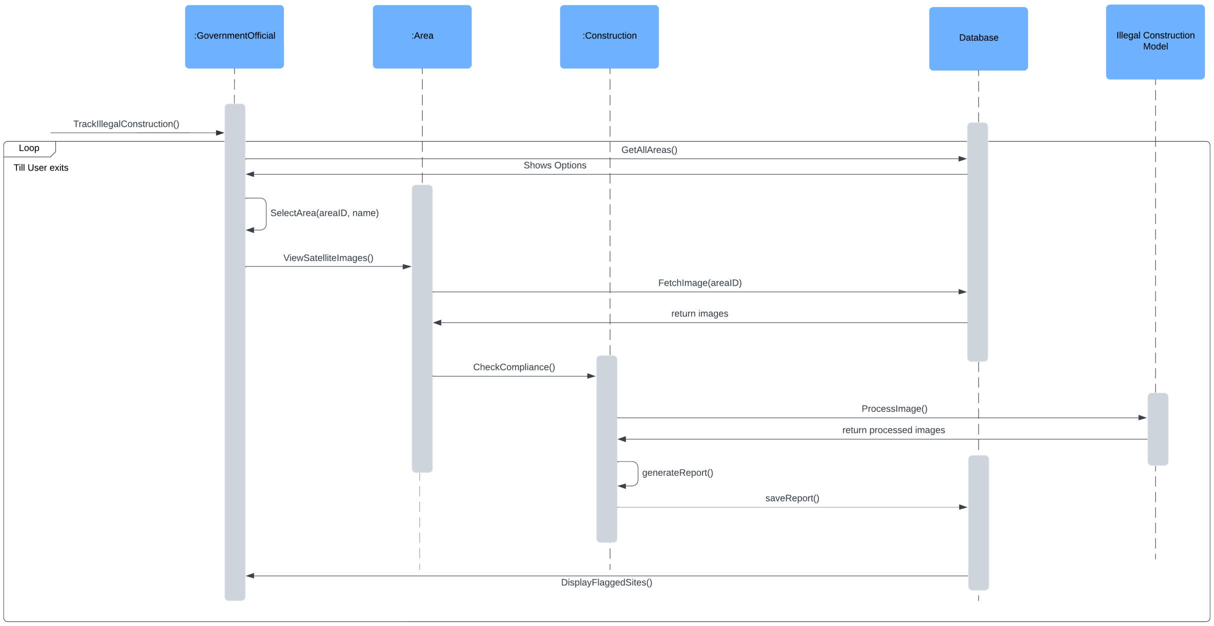

Figure 18: SD for Use Case 2: Detect Illegal Constructions

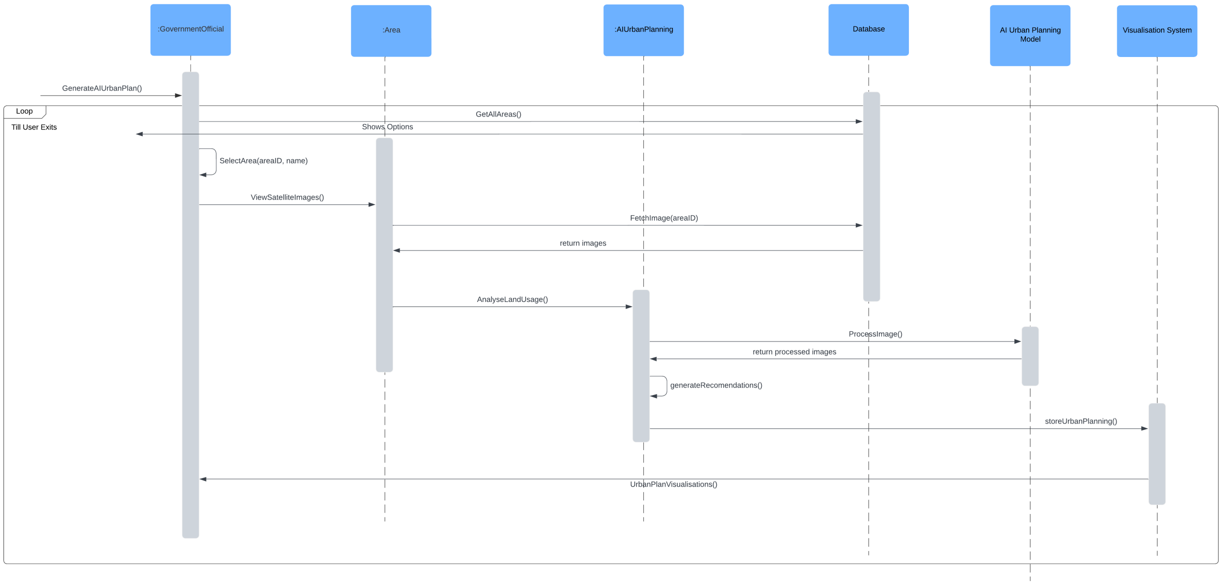

Figure 19: SD for Use Case 3: Generate an AI-driven Urban Plan

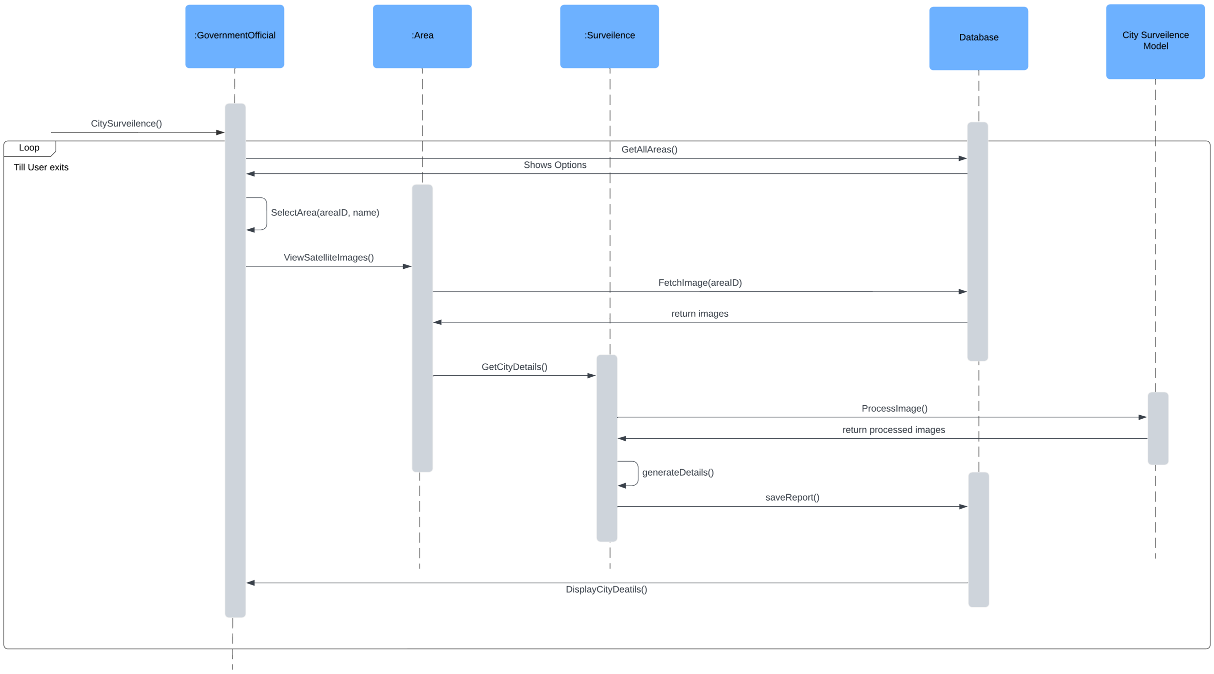

Figure 20: SD for Use Case 4: Request City Details

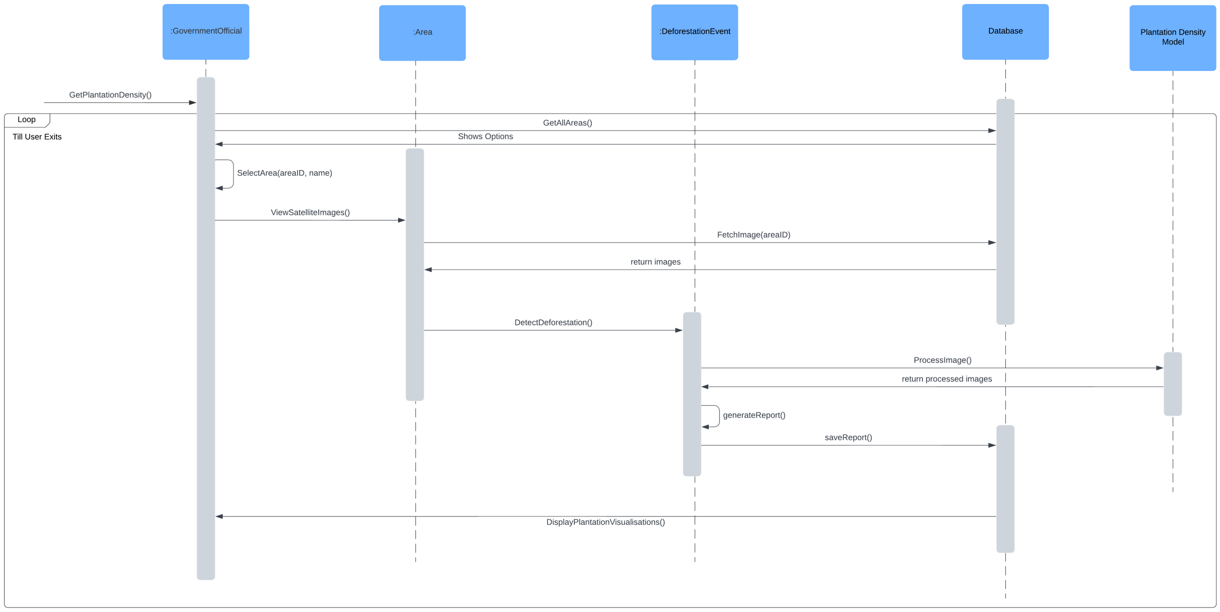

Figure 21: SD for Use Case 5: Access Plantation Density Visualisations

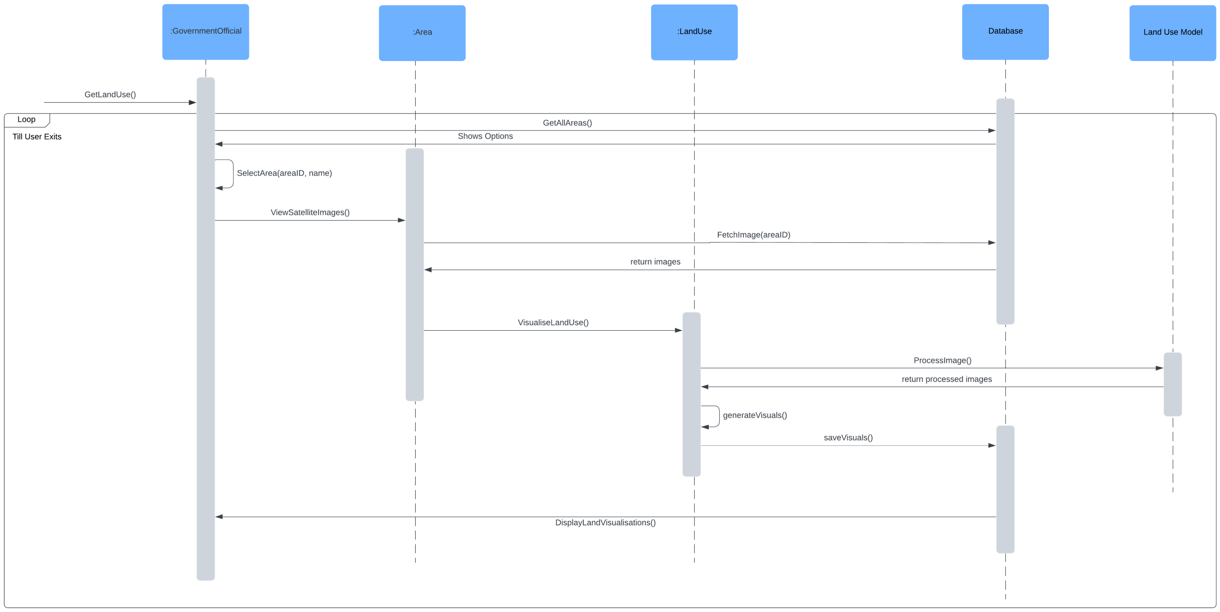

Figure 22: SD for Use Case 6: Access Land-Use Visualisations

12.6 ER Diagram

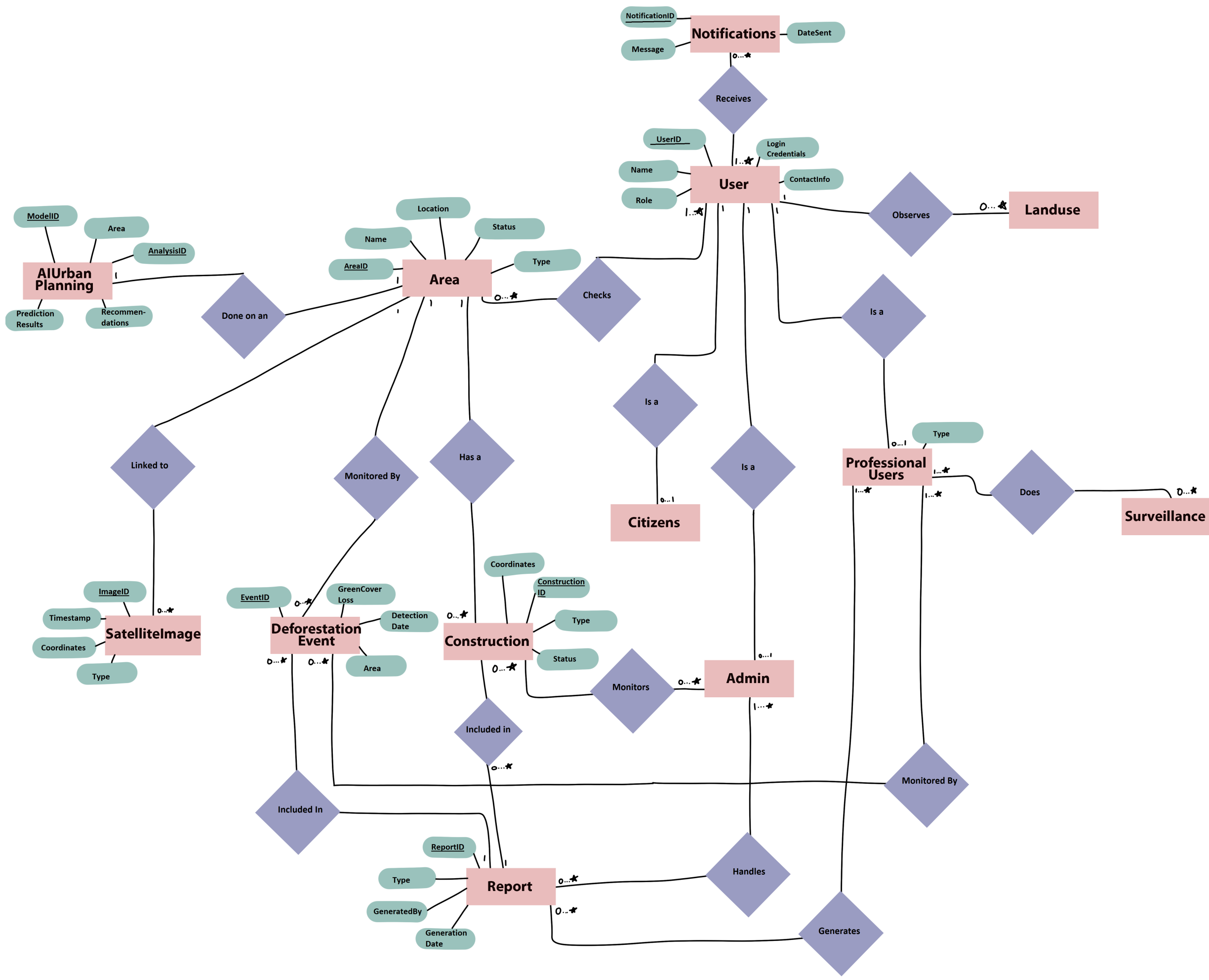

Figure 23: ERD - Entity Relationship Diagram

12.7 Workflow Diagram

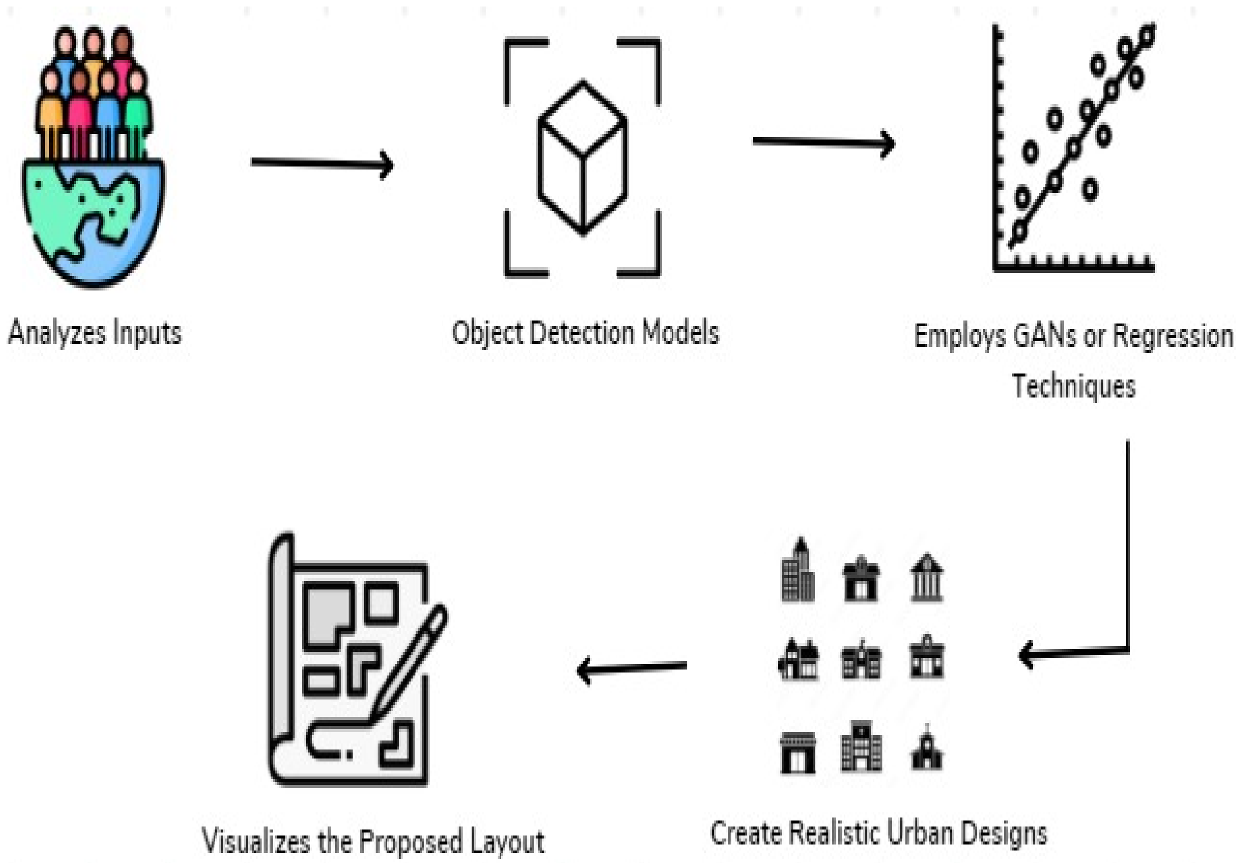

Figure 24: Workflow Diagram

13 Data Design

Data Design	Description
Data Storage and Processing	The back-end stores and processes user information, urban planning data, CDA construction rules and the information obtained from city surveillance. Data will be in different forms such as real-time satellite imagery and historical data that may be used for comparison.
Database	We will be using MongoDB as the primary database to store our user information, data for urban planning, plantation statistics, land usage data, and construction rules. The schema will be managed by `Mongoengine` for a structured data access. Redis may also be used as a caching solution for frequently accessed data.
Major Entities	- **Users**: Both citizens and government official information will be stored along with the user type that will help distinguish user access. Apart from that admin data will also be stored. - **Satellite Data**: Satellite images will be stored for city surveillance. - **Construction Data**: Information about illegal constructions, flagged areas, and history for the illegal constructions flagged previously. - **Environmental Data**: Plantation density and land usage statistics.
Data Processing	- **Google Earth Engine API** Will be used to fetch satellite images. - **AI Models**: AI algorithms process input data for generating urban plans and analyzing city patterns. - **Satellite Image Processing**: Satellite images will be processed to detect illegal constructions or environmental changes.
Visualization Data	- **Chart.js** Will be used for visualizing historical trends, such as deforestation rates and land use changes. - **Map Data**: Data related to geographical areas, constructions, and environmental analysis is visualized on Google Maps through the integration of the Google Maps API.
Reports and Notifications	- **Report Generation**: The system will generate detailed reports in PDF or CSV format about illegal constructions, land usage, and plantation density. - **Notifications**: Real-time push notifications, emails, or SMS alerts will be triggered for reporting illegal activities or for informing deforestation statistics.

Table 13: Data Design for the Urban Planning and Monitoring System

14 Conclusions

In this project, we will be developing an Urban Planning and a Surveillance System that will use satellite images and AI-driven analysis to flag illegal constructions and help find statistics regarding land usage and plantation density.

Key conclusions include:

1. **Integration of Technologies**: Successfully combines AI and GIS for informed decision-making.
2. **User-Centric Design**: Encourages active participation from citizens and officials by allowing them to send notifications and report any illegal constructions.
3. **Scalability**: The system must be designed for future enhancements and flexibility.

Future work will focus on improving AI models, incorporating user feedback, expanding data sets, developing a mobile application, and implementing the functionality for predictive analytics.

A Appendix 1

A.1 Definitions and Acronyms

CDA: Capital Development Authority - The agency responsible for approving city developments in Islamabad.

A.2 References

1. Capital Development Authority. (n.d.). By-laws & Regulations. CDA. https://www.cda.gov.pk/by-laws-regulations
2. Google Earth Engine. (n.d.). Google Research open buildings temporal v1 dataset. Google Developers. https://developers.google.com/earth-engine/datasets/catalog/GOOGLE_Research_open-buildings-temporal_v1
3. Google Earth Engine. (n.d.). JAXA ALOS PALSAR yearly FNF4 dataset. Google Developers. https://developers.google.com/earth-engine/datasets/catalog/JAXA_ALOS_PALSAR_YEARLY_FNF4